CERTIFIED BRACKETS AT THE PUBLISHED ZEFOZ POINTS OF ^{167}ER^{3+}:Y$_2$SIO$_5$: EXISTENCE, CURVATURE, AND THE MEASURED COST OF A COMPLETENESS CERTIFICATE

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Abstract. The magnetic-field points at which a hyperfine transition frequency of $^{167}\mathrm{Er}^{3+}$:$\mathrm{Y_2SiO_5}$ is stationary — the ZEFOZ (zero-first-order-Zeeman) points — set the predicted coherence times of a leading solid-state quantum-memory platform. The published atlas of these points (Matsuura et al., Phys. Rev. B **113**, 085421 (2026); arXiv:2412.10126) is produced by Newton iteration from finite grids of starting fields, and its authors state plainly that the number of points found depends on the initial grid. Two questions follow. First: do the published points survive rigorous scrutiny — does a true stationary point of the exact spin Hamiltonian provably exist near each tabulated field, and can its location, frequency, and curvature (the quantity that sets the predicted T_2) be pinned by two-sided bounds a skeptic can re-derive with a standard-library program? Second: can the published list be proven *complete* over an explicit field box at laptop scale? To the first question we give a full positive answer: for each of the twenty tabulated nonzero-field points of both crystallographic sites we certify, in exact rational arithmetic, eigenvalue and transition brackets of width 4×10^{-10} MHz, a gradient-norm bound $|\nabla f| \leq 3.2 \times 10^{-37}$ MHz/mT at an exactly specified rational field, two-sided brackets on all three eigenvalues of the 3×3 frequency Hessian, its signature, and — by a Krawczyk contraction with a rigorous third-derivative bound — the existence and local uniqueness of an exact stationary point within 2.9×10^{-14} mT of the stated field. The certified signatures show that none of the twenty published points is a local minimum of its transition frequency: thirteen are saddles and seven are local maxima. At zero field we certify exactly, by an explicit time-reversal identity, that all 120 transitions of both sites are stationary, with certified curvature brackets for the ten published zero-field pairs. To the second question the answer is a documented negative: a pre-registered kill condition for the completeness search fired. Certified branch-and-bound exclusion over the box $\|B\|_\infty \leq 100$ mT, measured on stratified sample chunks, gives a strict lower bound above 10^6 laptop-hours against a pre-registered budget of 200; the obstruction — quasi-degenerate hyperfine doublets that force per-level spectral-gap machinery below its validity radius — is quantified. Three errata in the reference paper are recorded, with version history: a load-bearing sign error in a printed quadrupole matrix that stood for eleven months (v1, v2; corrected upstream in v3), two tabulated frequencies inconsistent with their own stationary points by 2.7 and 4.3 MHz (certified restatements given), and an inconsistent sign pairing in a printed field vector. The certificate and a standard-library checker are the public unit.

1. The question

A hyperfine transition of an $^{167}\mathrm{Er}^{3+}$ ion in $\mathrm{Y_2SiO_5}$ has a frequency $f_{ij}(B) = \lambda_j(B) - \lambda_i(B)$, the difference of two eigenvalues of a 16×16 effective spin Hamiltonian depending on the applied magnetic field $B \in \mathbb{R}^3$. At a field where $\nabla f_{ij}(B) = 0$ — a ZEFOZ point [4] — the transition is first-order insensitive to magnetic noise, and the residual dephasing is set by the second-order response, i.e. by the 3×3 Hessian of f_{ij}. Operating at such points is the standard route to long spin coherence in rare-earth quantum memories [5, 6], and for $^{167}\mathrm{Er}^{3+}$:$\mathrm{Y_2SiO_5}$ — a rare-earth system combining a telecom-band optical interface with a nuclear spin — the atlas of ZEFOZ points and their curvatures published by Matsuura, Yasui, Kaji, Sasakura, Tawara, and Adachi [1] is the design input for spin-wave storage proposals [1, 7].

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That atlas is numerical in an essential way. The points are found by Newton iteration from finite grids of initial fields, and the authors state the consequence themselves: "because the total number of ZEFOZ points found depends on the initial grid conditions, the initial magnetic fields for the search were established by combining the three conditions below" [1, App. D]. No error bounds accompany the tabulated locations, frequencies, or curvatures; nothing excludes further points; and, as we found in the course of this work (Section 5), the printed inputs of the calculation carried a load-bearing sign error through two arXiv versions.

We therefore ask two questions, of different difficulty.

Question 1.1. Near each published ZEFOZ field, does a stationary point of the *exact* transition frequency provably exist? Can its location, its frequency, and the full spectrum of its Hessian be enclosed in two-sided brackets that are machine-checkable from first principles — exact rational arithmetic on the published Hamiltonian — by a program short enough to audit, with no trust in the software that produced the brackets?

Question 1.2. Can the published list be proven *complete* — every stationary point of every one of the 120 transitions enumerated — over an explicit field box, at laptop scale?

Question 1.1 we answer in full (Theorems 3.1 and 3.2). Question 1.2 carried a pre-registered kill condition, and the kill condition fired; Section 4 reports the measured exclusion statistics that fired it and the structural obstruction they expose. Both outcomes are results, and both are recorded with the same precision.

2. The model and its exact rationalization

The ground-manifold ($^4I_{15/2}$, Z_1) effective Hamiltonian of $^{167}\mathrm{Er}^{3+}$:$\mathrm{Y_2SiO_5}$ used throughout [1] is

$$H(B) \;=\; \mathbf{I}\cdot A\,\mathbf{S} \;+\; \mathbf{I}\cdot Q\,\mathbf{I} \;+\; \mu_B\, B\cdot g\,\mathbf{S} \;-\; \mu_N\, g_n\, B\cdot \mathbf{I}, \tag{1}$$

with electron spin $S = 1/2$, nuclear spin $I = 7/2$ (dimension 16), field B in the optical frame (D_1, D_2, b), and A, Q, g the site-dependent 3×3 matrices printed in [1, App. B] (v3), attributed there to Wang et al. [3]. We adopt every printed entry exactly as a rational number, take $\mu_B/h = 13.996244936$ MHz/mT and $\mu_N/h = 7.622593285\times 10^{-3}$ MHz/mT (CODATA truncations, again as exact rationals), $g_n = -0.1618$ as printed, and both crystallographic sites. The only irrationalities in $H(B)$ are the nuclear ladder amplitudes $\sqrt{7}$, $\sqrt{12}$, $\sqrt{15}$; the certificate encloses them in rational intervals validated by squaring, and all further arithmetic is exact rational rectangle-interval arithmetic with outward rounding. Levels $\lambda_0 \le \cdots \le \lambda_{15}$ are indexed ascending, $f_{ij} = \lambda_j - \lambda_i$, and all bracket claims below are claims about the exact spectrum of the exact matrix (1): every stated interval contains the corresponding true value. (Whether (1) with these matrices describes the physical crystal is a separate, experimental question; the certificate is unconditional about the mathematics and silent about the spectroscopy.)

3. Certified statements

3.1. Zero field.

Theorem 3.1 (exact zero-field stationarity)**.** *Let $M \in \{0, \pm 1\}^{16\times 16}$ be the signed permutation $M = R^{(1/2)} \otimes R^{(7/2)}$, $R^{(j)}|j,m\rangle = (-1)^{j-m}|j,-m\rangle$. Then, as an exact identity of matrices over $\mathbb{Q}[\sqrt{7},\sqrt{12},\sqrt{15}]$, for both sites,*

$$M\,\overline{H_0}\,M^{\mathsf{T}} = H_0, \qquad M\,\overline{Z_k}\,M^{\mathsf{T}} = -Z_k \quad (k = 1,2,3),$$

where $H_0 = H(0)$ and $Z_k = \partial H/\partial B_k$. Consequently $H(-B) = M H(B)^{\mathsf{T}} M^{\mathsf{T}}$, the spectrum of $H(B)$ is even in B; and since the certificate also proves the sixteen zero-field levels of each site *simple (disjoint brackets of width 2×10^{-10} MHz), every level is analytic near $B=0$ and*

$$\nabla f_{ij}(0)=0 \qquad \textit{exactly, for all } 0\le i<j\le 15, \textit{ both sites.}$$

This upgrades the folklore statement that "at zero field all transitions are ZEFOZ" to a machine-checked identity: the checker verifies the symbolic transform entry-by-entry in $\mathbb{Q}[\sqrt{d}]$ and the simplicity via Sylvester inertia counts (below). For the ten zero-field pairs tabulated in [1, Table 4] the certificate additionally carries two-sided brackets on all three eigenvalues of each Hessian $\nabla^2 f_{ij}(0)$ (Section 3.3); for example, for site 1 $(i,j)=(7,9)$ the certified Hessian spectrum is $\{-11.9905\ldots,\ +1.6899\ldots\times10^{-4},\ +1.1300\ldots\}$ MHz/mT^2, each bracket of width below 2.1×10^{-9}. The signatures are mixed: six of the ten published zero-field transitions have an indefinite Hessian, so at zero field those six frequencies are saddles in B; the remaining four — site 1 $(6,8)$ and $(6,9)$, site 2 $(6,8)$ and $(7,11)$ — carry the certified signature $(+,+,+)$, so their Hessians are positive-definite and at zero field those four frequencies are local minima in B.

3.2. The twenty published nonzero-field points.

Theorem 3.2 (certified brackets, curvature, and existence)**.** *For each of the twenty entries of* [1, Table 5] *(ten per site) there is an exactly specified rational field B^* (dyadic, denominator 2^{120}), within the rounding radius of the published coordinates, such that the certificate proves:*

- (i) *brackets of width 2×10^{-10} MHz for all sixteen $\lambda_n(B^*)$, hence a bracket of width 4×10^{-10} MHz for $f_{ij}(B^*)$;*
- (ii) $|\nabla f_{ij}(B^*)|\le 3.2\times10^{-37}$ *MHz/mT;*
- (iii) *two-sided brackets, of width at most 2.1×10^{-15} MHz/mT^2, for the three eigenvalues of $\nabla^2 f_{ij}(B^*)$, with certified signs;*
- (iv) *existence and uniqueness of a point $\hat{B}$ with $\nabla f_{ij}(\hat{B})=0$ in the box $B^*+[-r,r]^3$, $r=2^{-45}\,\mathrm{mT}\approx 2.9\times10^{-14}\,\mathrm{mT}$ (Krawczyk contraction; worst-case contraction ratio 0.096 over the twenty points).*

Corollary 3.3 (certified stationary-point types)**.** *Of the twenty published ZEFOZ points, none is a local minimum of its transition frequency. All ten site-1 points and the three site-2 points $(5,6)$, $(5,7)$, $(6,7)$ have certified signature $(-,-,+)$ (saddles); the remaining seven site-2 points, including the optimal $(14,15)$ transition of* [1]*, have signature $(-,-,-)$ (local maxima).*

The Hessian brackets are the certified form of the quantity the application consumes: in the noise model of [1, Eq. (8)] the predicted T_2 at a ZEFOZ point is a function of the second-order response along the noise distribution, i.e. of the enclosed Hessian; every curvature number a proposal takes from [1] can now be taken with a proof. For the site-2 $(14,15)$ point singled out in [1] the certified Hessian spectrum is $\{-1.65241\times10^{-4},\ -8.94878\times10^{-5},\ -1.44676\times10^{-6}\}$ MHz/mT^2 (brackets of width $<3\times10^{-16}$), and the certified field is $\hat{B}=(-378.9856,\ +73.2675,\ +502.3521)$ mT $\pm\,2.9\times10^{-14}$ mT — which also settles the sign question of Section 5, item (E3).

3.3. **Method, in brief.** All claims reduce to rational inequalities verified by the checker; the mathematical ingredients are classical. Eigenvalue brackets come from interval LDL^{T} factorizations of $H(B^*)-\mu I$ at certified shifts: when every pivot interval is sign-definite, Sylvester's law of inertia proves the number of eigenvalues below μ. High-precision eigenvectors (computed to 70 digits, then frozen as dyadic rationals with denominator 2^{170} — candidates, not trusted) are certified a posteriori: the exact interval residual $\|H\hat{v}-\tilde{\lambda}\hat{v}\|$ against the certified gaps gives Davis–Kahan $\sin\theta$ bounds [8] of order 10^{-42}; Hellmann–Feynman then turns them into gradient enclosures (ii), and second-order perturbation sums with certified matrix elements and certified spectral gaps give a signed enclosure of the full Hessian (iii), whose eigenvalues are bracketed by Gershgorin discs after an approximate rotation, with the exact Ostrowski congruence correction [11, Thm. 4.5.9] for the rotation's dyadic non-orthogonality. Existence (iv) is the Krawczyk test [9, 10]: with $C \approx (\nabla^2 f)^{-1}$ rational and $X = B^* + [-r, r]^3$,

$$K(X) = B^* - C\,\nabla f(B^*) + \big(I - C\,\nabla^2 f(X)\big)\,[-r, r]^3 \subset \operatorname{int} X$$

proves a unique zero of ∇f in X. The Hessian enclosure over X is the center enclosure inflated entrywise by $9\sqrt{3}\,Mr$, where $M = 192\,D^3(\gamma_i^{-2} + \gamma_j^{-2})$ is a rigorous bound on all third directional derivatives of f_{ij} over X, derived in Appendix A from a resolvent estimate and a Cauchy coefficient bound; D bounds $\sup_{|e|=1} \|\partial_e H\|_2$ and γ_i, γ_j are certified gap lower bounds over X. The margins are enormous — $|\nabla f(B^*)| \lesssim 10^{-37}$ against a needed $\lesssim 10^{-18}$ — because the flatness of clock transitions makes the Krawczyk test brutally stringent: the smallest certified Hessian eigenvalue magnitude among the twenty points is 5.67×10^{-8} MHz/mT^2, so the preconditioner norm reaches 1.8×10^7 and existence at these points is provable only from a candidate polished far beyond double precision. A Newton–Kantorovich argument with double-precision data alone would fail by many orders of magnitude; this appears to be intrinsic to certifying clock points, not an artifact of our route.

3.4. **Relation to the pilot certificate and to prior tools.** A same-program pilot release certified eigenvalue and transition brackets at these fields (without stationarity, curvature, or existence); its certificate re-verified here (checker exit 0; 352 brackets against an independent 60-digit diagonalization, zero containment failures). The present certificate strictly extends it. Generic verified eigensolvers exist — symveig [14] encloses spectra of fixed Hermitian matrices, and verified-numerics methodology is standard [13] — but we find no prior certified treatment of *parametric* eigenvalue-difference stationary points (gradients, Hessians, existence, or completeness) for spin Hamiltonians, and no certified object of any kind attached to a ZEFOZ table (novelty sweep of 2026-08-12, shipped with the release). The claims here are application-first in that sense; every mathematical tool is classical and credited above.

4. The completeness question: a pre-registered kill

The completeness target was specified in advance: certify that the published list is the whole list — every stationary point of every f_{ij}, both sites — over the box $\|B\|_\infty \le 100$ mT (the region the literature's own grids explored; note that all twenty published points lie *outside* this box, at 191 mT–5.3 T, so the completeness claim over the box would state that $B = 0$ is the only in-box stationary set). The pre-registered kill condition K2 read: at $\sim$20 laptop-hours of exclusion work, if the measured statistics project more than 200 laptop-hours to close the domain, kill the completeness claim and downgrade to certified existence and curvature at the known points.

The exclusion engine is the natural one: adaptive subdivision of the box; per box $X = B_c + [-r, r]^3$, certified eigenvalue brackets at B_c (Bauer–Fike residual discs with component counting), per-level gap bounds over X via the Lipschitz bound $\sqrt{3}\,rD$, Davis–Kahan eigenvector enclosures over X, certified Hellmann–Feynman gradient at B_c, a signed perturbation-sum Hessian enclosure over X, and the first-order exclusion test $|g_k(B_c)| > \sum_l \sup_X |\partial_k \partial_l f|\, r$ for some k. A box is closed when all 120 pairs are excluded; otherwise it splits.

The measured statistics are unambiguous. Certified per-box cost is 1.4–2.4 s (all 120 pairs amortized over one diagonalization). At box radius $r = 6.25$ mT and $r = 0.78$ mT, *zero* of the 120 pairs can be excluded at a generic center ($B_c = (50, 50, 50)$ mT): the certified Hessian sup-bounds, inflated by eigenvector drift $\propto rD/\gamma$, dominate the certified gradients. First majority exclusions appear only at $r \approx 0.1$ mT (87/120 at the generic center; 68/120 at $(6.25, 6.25, 6.25)$ mT), and 33–52 pairs survive even there. Adaptive runs on sample chunks (six chunks of half-width 0.39 mT, both sites, stratified in radius) give the following (all six exhausted their 340-s budgets with their subdivision trees unfinished):

site	chunk center (mT)	boxes	closed boxes	surviving pairs
1	generic $(50, 50, 50)$	234	0	2
1	near-origin $(1.2, 1.2, 1.2)$	176	0	32
1	far $(87.5, 62.5, 37.5)$	177	153	0
1	near-axis $(75, 5, 5)$	136	10	3
2	generic $(50, 50, 50)$	251	218	0
2	near-origin $(1.2, 1.2, 1.2)$	191	0	19

Closed boxes are finest-level ($r = 0.098$ mT) boxes with all 120 pairs excluded; surviving pairs are those still active at the depth cap. No chunk finished, so the certified cost per unit volume strictly exceeds each chunk's wall-clock/volume rate; the weakest measured rate is 713 s/mT^3. Scaling that weakest rate to the half domain (4×10^6 mT^3 after the $B \to -B$ symmetry) gives a strict lower bound of more than $\mathbf{7.9 \times 10^5}$ per site (1.6×10^6 for both sites) laptop-hours — before any refinement of the surviving pairs, which the calibration shows needs two further depth levels for the flattest pairs at generic fields and diverges for the near-origin web — against the pre-registered budget of 200. The lower bound alone exceeds the budget by a factor of 8,000, i.e. nearly four orders of magnitude. **K2 fired; the completeness claim is dead at laptop scale**, and this note's certificate is the pre-registered downgrade.

The obstruction is structural and worth recording. The zero-field spectrum of each site contains near-degenerate level pairs — certified splittings as small as 0.004 MHz (site 1) and 0.16 MHz (site 2), with six site-1 spacings below 36 MHz — alongside spacings of 300–900 MHz. Every per-level certified bound in the exclusion engine — eigenvector drift, matrix-element enclosures, perturbation denominators — degrades as the ratio rD/γ_n, with $D \approx 104$ MHz/mT the certified field-Lipschitz constant and γ_n the local gap of level n. A level inside a quasi-doublet of splitting γ is processable only for $r \lesssim \gamma/(4\sqrt{3}D)$, i.e. boxes of 10^{-5}–5×10^{-2} mT near the doublet-crossing set, which has positive two-dimensional measure in the box. Closing the domain therefore requires either cluster-projector enclosures (certifying the two-dimensional doublet subspace as a block, with block-resolved perturbation sums) or a different certification principle altogether; with single-level machinery the cost diverges near the crossing set. We record this as the concrete engineering frontier, not as an impossibility result.

5. Errata in the reference

Three defects of [1] were found and verified in the course of this work. We record them with version history; the first was corrected upstream by the authors, and we credit that correction.

(E1) *Site-1 quadrupole sign, v1 and v2.* Appendix B of v1 (2024-12-13) and v2 (2025-02-12) prints the site-1 Q matrix with $Q_{23} = Q_{32} = +15.5$ MHz. With $+15.5$ the paper's own zero-field frequencies (its Table 4) are irreproducible — off by up to 68 MHz — while with -15.5 they reproduce to ≤ 0.05 MHz. The paper's computations therefore used -15.5 throughout; the printed table was wrong for eleven months. v3 (2025-11-14), which matches the version of record [2], prints -15.5: corrected upstream, and any reader who rationalized v1/v2 as printed inherited a wrong Hamiltonian.

(E2) *Two tabulated frequencies inconsistent with their own points, v3.* In v3 Table 5, site 1, the pair $(6, 7)$ is tabulated at 745.8 MHz and $(4, 7)$ at 2216.2 MHz. The certified brackets at the stationary points nearest the stated coordinates (Theorem 3.2, existence certified) are $f_{67} = 748.5431883 \pm 2 \times 10^{-10}$ MHz and $f_{47} = 2220.5387344 \pm 2 \times 10^{-10}$ MHz: discrepancies of -2.74 and -4.34 MHz, roughly $50\times$ the rounding of the table. No alternative level pair at those fields is both stationary and at the tabulated frequency; the other eighteen entries agree with our brackets to ≤ 0.07 MHz (worst 0.0634 MHz, site 1 $(8, 13)$). The two values appear to be stale.

(E3) *Sign pairing of the highlighted site-2 field, v3.* The high-precision bullet for the site-2 $(14, 15)$ point prints $B_{(D_1,D_2,b)} = (\mp 378.9, \pm 73.2, \mp 502.3)$ mT alongside $B_{(B,\theta,\phi)} = (633.52 \text{ mT}, \mp 37.5383°, -10.9417°)$. The two are inconsistent: the stated angles (upper signs) give $(-379.0, +73.3, +502.3)$ mT, and the certified stationary point (Theorem 3.2) is at $(-378.99, +73.27, +502.35)$ mT. The printed Cartesian sign pairing should read $(\mp, \pm, \pm)$.

We verified (E1) against the v1, v2, and v3 source files directly and (E2), (E3) against certified brackets; the version of record [2] was not re-checked entry-by-entry against v3 beyond its Appendix B.

6. What is certified, what is enclosed, what is trusted

Certified (machine-checked from first principles by the standard-library checker): every claim of Theorems 3.1 and 3.2 and Corollary 3.3, as inequalities among exact rationals: the symbolic time-reversal identity, all inertia counts, all eigenvalue/transition brackets, the gradient enclosures, the signed Hessian enclosures and their eigenvalue brackets and signatures, and every Krawczyk contraction, including the third-derivative bound of Appendix A re-derived from the certified gaps. *Enclosed but pointing at published data:* the identification of each certified point with a row of [1, Table 5] is by proximity of the published rounded coordinates; the published rounding (0.001 T, 0.01°) is far coarser than our boxes. *Trusted:* the Python `fractions`/`json`/`math` standard library; the classical theorems cited (Sylvester, Davis–Kahan, Ostrowski, Krawczyk, and the resolvent/Cauchy lemma proved in the appendix); and nothing else — in particular no floating point, no eigensolver, and no code shared with the generator enters the verification. The kill-condition statistics of Section 4 are measurements of a private search engine and are *not* part of the certified surface; they are reported as run logs.

The computation, in one paragraph, per the program's protocol. Candidate fields and eigenvectors were produced by Newton iteration and 70-digit diagonalization (mpmath), then frozen as dyadic rationals; certification is a separate pass in exact rational interval arithmetic (outward rounding, denominator cap 2^{200}), ~ 4.6 s per point; the 23-object certificate (1.9 MB JSON: one time-reversal object, two zero-field objects, twenty Krawczyk objects) verifies in 39 s under `zefoz_checker2.py` (CPython 3.14, stdlib only, OS-independent), exit code 0; a six-item tamper battery (corrupted inertia count, shifted Hessian bracket, shifted gradient enclosure, flipped time-reversal sign, understated gradient norm, inflated Krawczyk radius) is rejected item-by-item with exit code 1, and an untampered control passes; the pilot certificate re-verifies (47 s, exit 0) and its 352 brackets contain an independent 60-digit recomputation. The branch-and-bound measurements ran as memory-capped queued jobs (34 min wall-clock total, peak 38 MB per job against the 2500 MB cap); all artifacts, the checker, the run logs, and the dated novelty sweep ship in the repository release; the search engine itself stays private per the program's certificate-plus-checker model.

7. Open questions

Question 7.1. Cluster-certified completeness. Can the box $\|B\|_\infty \leq 100$ mT be closed by replacing per-level Davis–Kahan enclosures with certified two-dimensional doublet-projector enclosures (block perturbation sums with inter-cluster gaps only), and what is the measured cost? The engine interface already isolates the per-level bounds; this is the one concrete route our measurements leave open at laptop scale.

Question 7.2. Existence certificates from double precision. The Krawczyk test at clock points needed candidates polished to 10^{-40}; is there a certification principle (exploiting the evenness of the spectrum in B, or the analyticity of f_{ij}) whose stringency does not scale with the inverse Hessian, i.e. that certifies flat stationary points from double-precision data?

Question 7.3. The physical atlas. The certified signatures (all twenty published nonzero-field points saddles or maxima, none minima; six of the ten published zero-field pairs saddles, the other four local minima) and the certified zero-field curvatures are properties of the model (1) with the printed matrices. Do measured second-order shifts at the $(14, 15)$ site-2 point distinguish the certified Hessian spectrum $\{-1.652 \times 10^{-4}, -8.949 \times 10^{-5}, -1.447 \times 10^{-6}\}$ MHz/mT^2 from the scalar $|S_2|$ summaries in use?

Question 7.4. Which certified object does the experiment want next: brackets at the $|B| \leq 3$ T points of the v3 search for site-independent field orientations, certified *optimality* of the $(10, 11)/(14, 15)$ choices within the tabulated list, or completeness over the small ball $|B| \leq 25$ mT actually swept by the published grids?

Appendix A. A third-derivative bound from a resolvent estimate

Lemma A.1. *Let H be Hermitian, λ a simple eigenvalue with $\gamma = \operatorname{dist}(\lambda, \operatorname{spec} H \setminus \{\lambda\}) > 0$, V Hermitian with $\|V\| \leq D$. For complex t with $|t| < \rho := \gamma/(4D)$ the matrix $H + tV$ has exactly one eigenvalue $\lambda(t)$ in the open disc of radius $\gamma/2$ about λ; $\lambda(t)$ is analytic, and $\left|\lambda'''(0)\right| \leq 192\, D^3/\gamma^2$.*

Proof. This is the standard resolvent perturbation argument (cf. [12]) with constants tracked. For z on the circle Γ of radius $\gamma/2$ about λ, $\|(H - z)^{-1}\| = \operatorname{dist}(z, \operatorname{spec} H)^{-1} \leq 2/\gamma$, so for $|t| < \rho$ the Neumann series of $(H + tV - z)^{-1} = (H - z)^{-1} \sum_{k \geq 0} \big(-tV(H - z)^{-1}\big)^k$ converges ($\|tV(H - z)^{-1}\| \leq \rho D \cdot 2/\gamma = 1/2$). Hence Γ lies in the resolvent set of $H + tV$ for all such t; the Riesz projection $P(t) = -\frac{1}{2\pi i} \oint_\Gamma (H + tV - z)^{-1} dz$ is analytic in t and of constant rank 1, and $\lambda(t) = \operatorname{tr}\big[(H + tV)P(t)\big]$ is the unique eigenvalue inside Γ, analytic in t, with $|\lambda(t) - \lambda| \leq \gamma/2$. Writing $\lambda(t) - \lambda = \sum_{k \geq 1} a_k t^k$ on $|t| < \rho$ and applying the Cauchy estimates on circles of radius $\rho' \uparrow \rho$, $|a_3| \leq (\gamma/2)/\rho^3$, so $|\lambda'''(0)| = 6|a_3| \leq 3\gamma\,(4D/\gamma)^3 = 192\, D^3/\gamma^2$. □

Applied at every real point B of a box X with $V = \partial_e H = \sum_k e_k Z_k$ ($|e| = 1$, so $\|V\| \leq D := (\sum_k \|Z_k\|_2^2)^{1/2}$) and with γ replaced by certified box-uniform gap lower bounds $\tilde{\gamma}_i, \tilde{\gamma}_j$, the lemma gives, for the cubic form $g_B(e) = \partial_e^3 f_{ij}(B)$, $\sup_{|e|=1} |g_B(e)| \leq M := 192\, D^3(\tilde{\gamma}_i^{-2} + \tilde{\gamma}_j^{-2})$. Polarization of the symmetric trilinear form $T = D^3 f_{ij}(B)$, $T[a, b, c] = \frac{1}{48} \sum_{\varepsilon \in \{\pm 1\}^3} \varepsilon_1 \varepsilon_2 \varepsilon_3\, g_B(\varepsilon_1 a + \varepsilon_2 b + \varepsilon_3 c)$, with $\|\varepsilon_1 a + \varepsilon_2 b + \varepsilon_3 c\| \leq 3$ for unit a, b, c, gives $|T[a, b, c]| \leq \frac{8 \cdot 27}{48} M = 4.5\, M$; the certificate uses the laxer constant $9M$. Integrating along the segment from B^* to $B \in X$ bounds each Hessian entry's variation by $9M\|B - B^*\| \leq 9\sqrt{3}\, Mr$, which is the entrywise inflation used in the Krawczyk operator, and which the checker recomputes from the certified D and gaps.

References

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